

# Using Double Machine Learning to Estimate the Effect of Pre-Primary Education on Student Achievement in Germany

Evidence from PISA 2022

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**Github Repository:**

<https://github.com/julianlohmann/CausalMLproject>

# Analytical Framework: Causal Effect of Pre-primary School Education

## Data & Variables

### General data description

**Data:** German subsample of PISA 2022 representative sample of all 15-year-old students

**Sample:** Complete-case sample ( $N = 1,731$ )

$N_{\text{no pre-schooling}} = 803$ ;  $N_{\text{pre-schooling}} = 928$ .

### Variable overview

**Treatment ( $T$ ):**

**Pre-primary education:** binary

**Outcome ( $Y$ ):**

**Reading ability:** continuous

**Covariates ( $X$ ):**

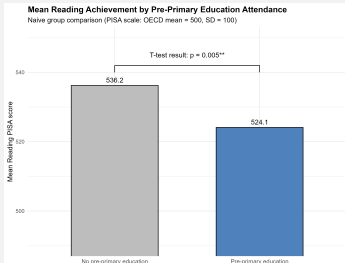
**Student:** gender, age, migration

**Parents:** SES, education, attitudes, migration

**Collider ( $Z$ ):**

A lot of potential colliders and mediators in the dataset - but hopefully all were excluded from the covariate list.

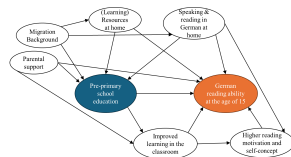
## T-Test Baseline



## Naïve conclusion

Protect your children from pre-primary school education!

## Structural Model



## First Naïve Findings

T-test implies that pre-primary education reduces reading ability by around 12 PISA points on average.

**This result may be confounded,**  
motivating a reanalysis with Double ML